STA141B-HW1-Functions

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Assignment 1 (Functions)

[1] “C:/Users/stela/Downloads/UCD Class/STA141B/offline.gz”

[1] “# timestamp=2006-02-11 08:31:58” [1] “t=1139643118744;id=00:02:2D:21:0F:33;pos=0.0,0.0,0.0;degree=0.0;00:14:bf:b1:9
38,2437000000,3;00:0f:a3:39:e1:c0=-54,2462000000,3;00:14:bf:b1:97:90=-56,2427000000,3;00:14:bf:3b:c7:c6=-
67,2432000000,3;00:14:bf:b1:97:81=-66,2422000000,3;00:14:bf:b1:97:8d=-70,2442000000,3;00:0f:a3:39:e0:4b=-
79,2462000000,3;00:0f:a3:39:dd:cd=-73,2412000000,3;00:0f:a3:39:e2:10=-83,2437000000,3;02:00:42:55:31:00=-
85,2457000000,1”

Appendix: R code

```
knitr::opts_chunk$set(echo = TRUE)
set.seed(1)
ff <- list.files("C:/Users/stela/Downloads/UCD Class/STA141B/",pattern="\\.gz$",full.names = TRUE)
ff[1]
offline_data = readLines(ff[1])
# check to understand data strcture
offline_data[1]
offline_data[5]
#function
readWirelessDataset=function(path) {
  x=readLines(gzfile(path))

  #empty for storing
  all_rows=list()
  all_cols=c()

  parsed_rows=list()

  #loop for each scan obs
  for (i in seq_along(x)){

    #if empty, skip
    if (nchar(x[i])==0) next

    #split by ; between elements
    parts <- strsplit(x[i], ";")[[1]]
    #split by = between value title and value content
    kv <- strsplit(parts,"=", fixed=TRUE)

    #take out value titles and content as characters
    keys <- sapply(kv, function(z) z[1])
    vals <- sapply(kv, function(z) if (length(z) > 1) as.character(z[2]) else NA_character_)

    #skip line if missing time and id
    if (!any(keys=="t")||!any(keys=="id")) next

    #create base while suppressing warning
    row=list(
      time   = suppressWarnings(as.numeric(vals[keys == "t"])),
      id     = vals[keys == "id"],
      pos    = vals[keys == "pos"],
      degree = suppressWarnings(as.numeric(vals[keys == "degree"]))
    )

    #the rest is for wifi signal, mac ID
    mac_idx <- !(keys %in% c("t", "id", "pos", "degree"))

    if (any(mac_idx)) {

      #take out value titles and content
      mac_keys <- keys[mac_idx]
```

```r
      mac_vals <- vals[mac_idx]

      #loop for each entry
      for (j in seq_along(mac_keys)) {

        #create column name
        col_name <- paste0("signal_", mac_keys[j])

        #convert content to character string
        val=as.character(mac_vals[j])

        #split by ,
        #convert to numerical, suppress warning, assign NA if empty
        if (!is.na(val) && val != "") {
          signal_parts <- strsplit(val, ",")[[1]]
          row[[col_name]] <- suppressWarnings(as.numeric(signal_parts[1]))
        } else {
          row[[col_name]]=NA
        }
        all_cols <- unique(c(all_cols, col_name))
      }
    }

    #store all rows to list
    parsed_rows[[length(parsed_rows) + 1]]=row
  }

  #combine into data frame
  df <- do.call(rbind, lapply(parsed_rows, function(r) {
    #assign NA if row missing a column
    for (col in all_cols) {
      if (!col %in% names(r)) r[[col]]=NA
    }
    as.data.frame(r, stringsAsFactors=FALSE)
  }))

  #clean row names
  rownames(df)=NULL

  return(df)
}
cat("\\newpage") #new page
```